

1) Find the volume generated by rotating the region bounded by the curves around the y -axis. The curves are $y = \frac{1}{x}$, $y = 0$, $x = 1$, and $x = 4$. Use cylindrical shells.

$$V = 2\pi \int_a^b (r)(h) dx$$

$$V = 2\pi \int_1^4 x \left(\frac{1}{x}\right) dx$$

$$V = 2\pi \int_1^4 1 dx$$

$$V = 2\pi \left[x \right]_1^4$$

$$2\pi(4-1)$$

$$2\pi(3)$$

$$(6\pi)$$

2) Find the volume generated by rotating the region bounded by the curves around the y -axis. The curves are $y = x^3 + x$, $y = 0$, $x = 0$, and $x = 1$. Use cylindrical shells.

$$V = 2\pi \int_a^b (r)(h) dx$$

$$V = 2\pi \int_0^1 (x)(x^3) dx$$

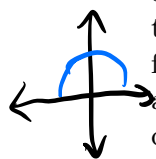
$$x^4 dx$$

$$V = 2\pi \left[\frac{x^5}{5} \right]_0^1$$

$$2\pi \left(\frac{1}{5} - 0 \right)$$

$$\frac{2\pi}{5}$$

3) Consider the region bounded by the x -axis and the equation $y = \sqrt{r^2 - x^2}$. Rotate this region about the x -axis and identify the shape created. Then use either the shells method or the disk/washers method find the volume and compare to common mathematical knowledge. Decide which technique is more appropriate. (Hint: you know this shape already, just get there conceptually and let the calculus catch up to you.)



$$V = \pi \int_{-r}^r \pi(r^2 - x^2) dx$$

$$V = 2 \left(\pi \int_0^r \pi(r^2 - x^2) dx \right)$$

$$V = 2\pi \left[r^2 x - \frac{x^3}{3} \right]_0^r$$

$$V = 2\pi \left(r^3 - \frac{r^3}{3} \right)$$

$$2\pi \left(\frac{2}{3} r^3 \right)$$

$$\frac{4}{3} \pi r^3$$

more appropriate
less BS

$$y = \sqrt{r^2 - x^2}$$

$$x^2 = r^2 - y^2$$

$$x = \pm \sqrt{r^2 - y^2}$$

$$V = 2\pi \int_0^r (2)(y) dy$$

$$V = 2\pi \int_0^r y(2\sqrt{r^2 - y^2}) dy$$

$$= 4\pi \int_0^r y\sqrt{r^2 - y^2} dy$$

$$u = r^2 - y^2 \quad du = -2y dy$$

$$y dy = -\frac{1}{2} du$$

$$4\pi \int_{r^2}^0 \sqrt{u} \left(-\frac{1}{2}\right) du = 2\pi \int_0^{r^2} u^{\frac{1}{2}} du$$

$$V = 2\pi \left[\frac{2}{3} u^{\frac{3}{2}} \right]_0^{r^2}$$

$$2\pi \left(\frac{2}{3} (r^2)^{\frac{3}{2}} \right)$$

$$(r^2)^{\frac{3}{2}}$$

$$2\pi \frac{2}{3} r^3$$

$$\frac{4}{3} \pi r^3$$

4) Find the volume generated by wrapping the area bounded by $y = x^2$ and $y = 4x - x^2$ about the line $x = -3$ using either shells or disks/washers.

$$x^2 = 4x - x^2$$

$$2x^2 - 4x = 0$$

$$2x(x-2)$$

$$V = 2\pi \int_a^b (r)(h) dx$$

$$r(x) = x - (-3) = x+3$$

$$V = 2\pi \int_0^2 (x+3)(4x-2x^2) dx$$

$$\begin{aligned} & \downarrow \begin{aligned} & 4x^2 + 12x - 2x^3 - 6x^2 \\ & (-2x^3 - 2x^2 + 12x) \end{aligned} \\ & 2\pi \int_0^2 2(-x^3 - x^2 + 6x) dx \end{aligned}$$

$$4\pi \left[-\frac{x^4}{4} - \frac{x^3}{3} + 3x^2 \right]$$

$$h(x) = (4x - x^2) - x^2$$

$$4x - 2x^2$$

$$\int_0^2 \left(-\frac{2^4}{4} - \frac{2^3}{3} + 3(2) \right) dx$$

$$-4 - \frac{8}{3} + 12$$

$$8 - \frac{8}{3}$$

$$\frac{16}{3}$$

$$V = 4\pi \left(\frac{16}{3} \right)$$

$$\frac{64\pi}{3}$$

5) Find the arc length of the curve $y = -4x + 1$ from $x = -1$ to $x = 1$ by first using the Pythagorean Theorem and then by using integration. Your answers should agree.

$$a^2 + b^2 = c^2$$

$$y = -4(-1) + 1$$

$$(-1, 5)$$

$$\Delta x = 2 \quad \Delta y = -8$$

$$L = \sqrt{\Delta x^2 + \Delta y^2}$$

$$L = \sqrt{4 + 64}$$

$$\sqrt{68}$$

$$(\text{google} \rightarrow 2\sqrt{17})$$

$$y = -4(1) + 1$$

$$(1, -3)$$

arc length

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = \frac{d}{dx}(-4x + 1) = -4$$

$$L = \int_{-1}^1 \sqrt{1 + (-4)^2} dx$$

$$\int_{-1}^1 \sqrt{1 + 16} dx$$

$$\int_{-1}^1 \sqrt{17} dx$$

$$L = [\sqrt{17}x]_{-1}^1 = \sqrt{17}(1) - \sqrt{17}(-1) = 2\sqrt{17}$$

6) Use calculus to find the arc length of the curve $y = \sqrt{4 - x^2}$ from $x = -2$ to $x = 2$.

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$y = (4 - x^2)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2} (4 - x^2)^{-\frac{1}{2}} (-2x)$$

$$= -\frac{x}{\sqrt{4 - x^2}}$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\sqrt{\frac{4}{4 - x^2}}$$

$$\frac{2}{\sqrt{4 - x^2}}$$

$$L = \int_{-2}^2 \frac{2}{\sqrt{4 - x^2}} dx$$

$$L = 4 \int_0^2 \frac{1}{\sqrt{4 - x^2}} dx$$

$$1 + \left(\frac{x^2}{4 - x^2}\right)$$

$$\frac{(4 - x^2) + x^2}{4 - x^2}$$

$$\frac{4}{4 - x^2}$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a}$$

$$L = 4 \lim_{t \rightarrow 2^-} \left[\arcsin\left(\frac{x}{a}\right) \right]_0^t$$

$$L = 4(\arcsin(1) - \arcsin(0))$$

$$4\left(\frac{\pi}{2} - 0\right)$$

$$\frac{4\pi}{2}$$

$$(2\pi)$$

7) Set up (but do not solve) the integral that would give the surface area of a solid created by rotating $y = \sqrt{x^2 - 2}$ around the x -axis from $x = 2$ to $x = 6$. Once your integral is set up, approximate its value by using technology.

$$S = \int_2^6 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = \frac{d}{dx} [(x^2 - 2)^{\frac{1}{2}}]$$

$$\frac{1}{2} (x^2 - 2)^{-\frac{1}{2}} (2x)$$

$$\frac{x}{\sqrt{x^2 - 2}}$$

$$\sqrt{1 + \left(\frac{x}{\sqrt{x^2 - 2}}\right)^2}$$

$$y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{x^2 - 2} \cdot \frac{\sqrt{2x^2 - 2}}{\sqrt{x^2 - 2}}$$

$$\sqrt{2x^2 - 2}$$

$$S = 2\pi \int_2^6 \sqrt{2x^2 - 2} dx$$

$$\approx 137.1582$$

8) Set up (but do not solve) the integral that would give the surface area of a solid created by rotating $x = \sin(y)$ around the y -axis from $y = 0$ to $y = \pi$. Once your integral is set up, approximate its value by using technology.

Wompwomp

9) Mixed review time! Compute **both** the integral and derivative of each of the following four functions. For derivatives, product and chain rules will be needed. For integrals you will only need u -substitution techniques and basic facts.

$$f(x) = e^{3x}$$

$$g(x) = xe^{3x^2}$$

$$h(x) = x^2e^{3x^3}$$

$$j(x) = \sin(x)e^{\cos(x)}$$

10) Mixed review time! Compute the following integral by setting $u = x^2 + 1$. Please note that this will be one of (if not) the last times I will provide the choice for u . If you are still struggling to see the optimal choice please reach out so we can drill them for twenty minutes or until you feel more confident.

$$\int \frac{x^3}{x^2 + 1} dx$$